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Introduction

For our first analysis, we looked at NASA's dataset that collected signals after observing around 150,000 stars. Each signal comes from dips in brightness that occur when a planet passes in front of the star, which they labeled as a Kepler Object of Interest (KOI). The catch being that most of the KOI's turn out not to be planets. It could be the result of two stars eclipsing each other, a brighter neighboring star, or instrument noise that can all fake a signal. The follow up observations are expensive just to rule them out. Using NASA's table of roughly 7328 KOI's that have already been settled as confirmed, false positives and compared four classification methods. Logistic regression, linear SVM, polynomial SVM, and RBF SVM using the kernel trick, with the RBF SVM performing the best with an F1 score of 0.898 compared to 0.812 for logistic regression.

Our second analysis was performed on data collected from the MAGIC Gamma Telescope. Ground-based gamma-ray telescopes detect high-energy particles by observing the brief flashes of light produced when they interact with Earth's atmosphere. However, these observations also include hadron events, which create similar signals and act as background noise. Separating gamma-ray events from hadron events is an important step for astronomers attempting to identify distant light sources or celestial objects, given the time, cost, and energy required to make these identifications astronomically large. Our analysis goes into how well machine learning can automate this classification process using data collected from the MAGIC Gamma Telescope. We compare logistic regression and several (SVM) models to determine which most effectively distinguishes between the two particle classes. Typically RBF SVM models performed best, with a cross-validated performance of 0.846 at the expense of doubling runtime for training.

Repository:https://github.com/Zamaughan/cs5830_project7
Slides: Presentation

Dataset 1

We used the KOI table from NASA Exoplanet Archive, which has 9564 KOIs and 49 columns. Our target is koi.disposition, which says that a KOI is confirmed, a false positive, or still a candidate. We kept only the first two since the candidates did not have a known answer yet. We also used 12 explanatory variables in three groups. Group 1: the shape of the dip itself, being the orbital period, duration, depth, impact parameter, and signal to noise. Group 2: the host star consisting of temperature, surface gravity, radius and brightness. Group 3 consists of the values NASA calculated from those, planet radius, equilibrium temperature, and how much starlight the planet receives.

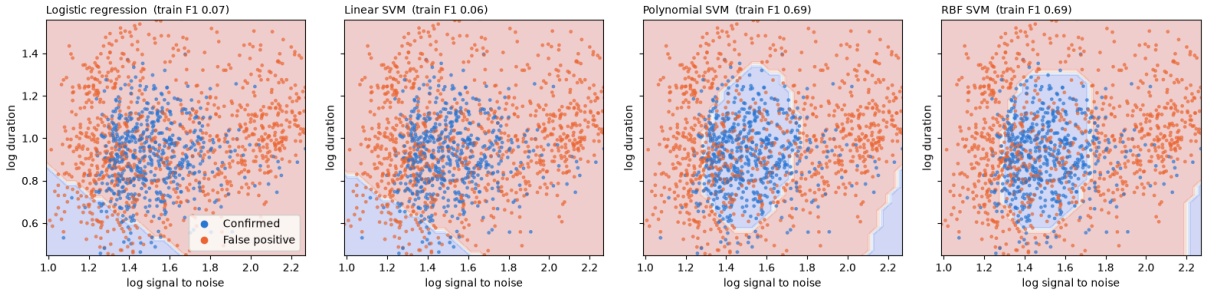
Analysis Technique 1

Logistic regression and SVMs looked to be a great fit for this since we're trying to predict a category based from numeric measurements. While logistic regression is fast and simple it

ended up being limiting since can only create a boundary in a straight line, but it provides a great baseline. Since we can have a multitude of false positives, having the ability to create curved boundaries using SVMs with polynomial or RBF allows us to be more flexible in our boundaries.

Results 1

Comparing the four models on 1457 KOIs we held back for testing, always guessing false positive was right 65.5% of the time on this test set. The two straight line models were nearly identical, logistic regression and linear SVM scoring 0.812 and 0.809, respectively. Both curved SVMs did better with polynomial SVM at 0.889 and RBF at 0.898.

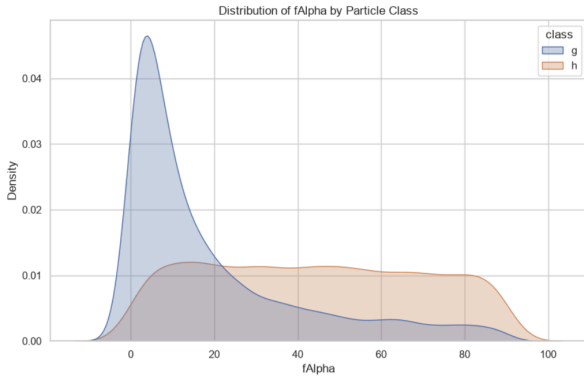


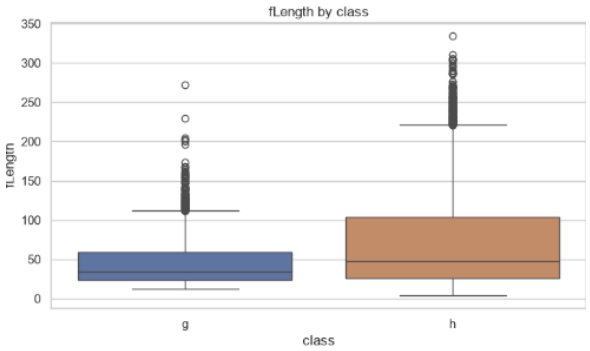
Technical 1

That NASA csv starts with comment lines, which we skipped. Dropped 363 labeled KOIs with no fitted transit model because missing values showed up almost entirely in false positives. We left out columns produced by NASA's own vetting; koi_pdisposition, koi_score, and the four koi_fpflag. Applied log1p to nine features that span several orders of magnitude and standard scaling inside an sklearn pipeline. Roughly 614 stars host more than one KOI and those KOIs share identical star values; we split them 80/20 by star, so stars appear in both the training and test splits.

Dataset 2 - MAGIC Gamma Telescope

The MAGIC Gamma Telescope dataset contains 19020 observations, each described by 10 numerical features of the resulting atmospheric shower. These measurements characterize properties such as the length, width, orientation, and concentration of the observed light distribution. The target variable identifies each event as either a gamma-ray event (g) or a hadron event (h). The dataset contains 12332 gamma-ray observations and 6688 hadron observations which creates a significant class imbalance. All of our explanatory variables were quantitative which made the dataset well suited to logistic regression and SVM classification. The best variables determined for the evaluation were fAlpha and fLength, which makes sense given that a gamma-ray event will be shorter and brighter while background hadron events will be longer and less noticeable. Attempting to plot these points as pairs was difficult because of the class imbalance, but individually you can see where the variables are distinct between classes.



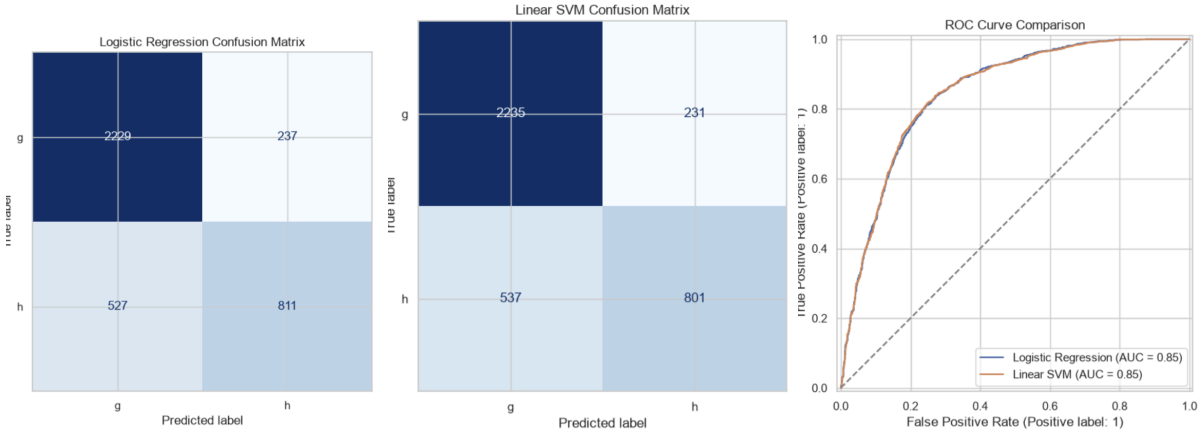


Analysis Technique 2

We evaluated logistic regression and three types of SVMs: linear, polynomial, and radial basis function (RBF). These methods were selected because they can classify observations using numerical measurements and provide different approaches to separating the two classes. Logistic regression and linear SVMs establish a baseline using linear decision boundaries, while polynomial and RBF SVMs can capture more complicated, nonlinear relationships. All models used standardized values for the variables to ensure that the differences in measurement scales did not disproportionately influence predictions. We also investigated balanced class weights to determine whether accounting for the unequal class frequencies could improve the identification of hadron events.

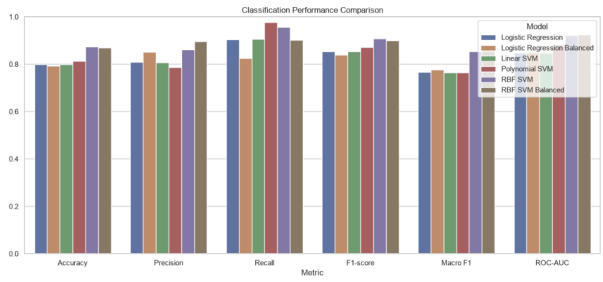
Results 2

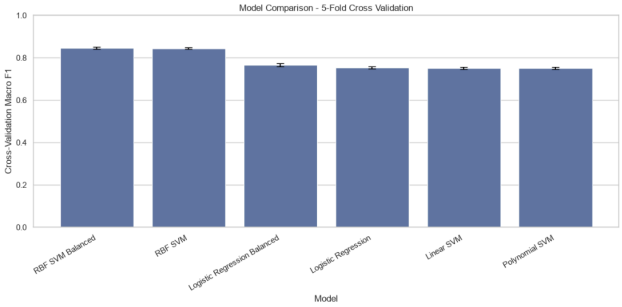
The initial logistic regression and linear SVM models achieved approximately 80% accuracy, with ROC-AUC scores around 0.85. Their similar performance suggests that the linear decision boundaries capture some of the distinguishing characteristics of the two particle classes, but there's still room for improvement. The polynomial SVM achieved 81.3% accuracy, while the RBF SVM achieved 87.3% accuracy and a ROC-AUC score of 0.922. The RBF model also achieved a gamma-ray recall of 96%. Introducing balanced class weights improved its ability to recognize hadron events, increasing hadron recall from 72% to 81%, while gamma-ray recall decreased from 96% to 90%.



The balanced RBF SVM model achieved the highest average macro F1-score when performing five-fold cross-validation, which barely surpassed the unweighted RBF model. Ultimately, the

RBF SVM provides the strongest overall classification approach while class weighting offers a useful alternative when correctly identifying background hadron events is particularly important to avoid overspending to confirm. The aforementioned highest average macro F1-score being 0.846 for the balanced RBF SVM compared to 0.845 for the unweighted model and 0.765 for the balanced logistic regression model. The time to train, however, doubled from 1.02 seconds for the logistic regression models to nearly 2.14 seconds for the RBF SVMs.





Technical 2

No missing observations within the data were identified in the dataset. Logistic regression and Linear SVM were initially evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Because the initial models showed relatively poor hadron event recall, balanced class weights were investigated. Polynomial and RBF kernels were subsequently introduced to capture nonlinear decision boundaries, with the RBF kernel providing the largest improvement particularly when class weights were balanced. Five-fold stratified cross-validation was performed on the training data to assess model consistency. The balanced RBF SVM achieved the highest cross-validated macro F1-score, although the unweighted RBF model achieved slightly higher average accuracy. Hyperparameters were tested but rather than exhaustively optimized, leaving further tuning as a potential avenue for future model improvement.